intended to convey his positive sentiments. This results in an artefact/effect, which is the review document with a positive sentiment. Given this document/effect, the task is to predict what the underlying cause/sentiment contained in the document is — essentially, predicting the cause (the sentiment which the user wanted to convey) from the effect/artefact, namely, the review document.

Given a document, the NLP systems are not learning what constitutes the sentiment of a document. Instead, given a set of labelled documents in the training data, they are trying to learn what features are predictive of the label associated with each review document for that training set. They do not have the knowledge to differentiate between features that actually carry semantic signals expressing sentiments and features that are spurious statistical patterns appearing in the training data. Their task is to predict the label, based on the given set of documents, using whatever features, heuristics or shortcuts they can learn from the training data.

To make this idea concrete, think of a sentiment analysis data set, where all references to Donald Trump are negatively labelled sentences. The model ends up learning that the word 'Trump' is highly associated with a negative sentiment. If it is shown a test sentence containing the name 'Trump', such as, "This story is just like Trump," it is likely to label it as negative in the absence of any other sentiment-bearing words.

Consider our earlier example, "A great white shark bit me on the beach." The NLP model is likely to have seen the words 'great' and 'white' in positive sentiment-bearing sentences. Hence the model associates these words blindly as predictive of a positive sentiment. Given the infrequent occurrence of the word 'bit' as a verb, the model may not be able to attribute a sentiment to it. So it ends up using the spurious pattern 'great white' to wrongly associate a positive label with this sentence.

This happens because of the associative learning paradigm. Just because a certain set of features that the model has identified as predictive of a specific label are present in a given document, the model associates that label with the document. The popular statement, "Correlation is not causation," warns against such mis-predictions. Just because the words 'great white' are often present in positive sentiment-bearing documents, one cannot blindly assume that these words are always present because of the underlying positive sentiment. In the above example, the words 'great white' were used as adjectives to describe a specific variety of shark, and did not carry a positive sentiment for the sentence. The NLP model is just not able to distinguish between relevant features and irrelevant features, given a specific context, in the associative learning paradigm. This makes it brittle.

Now, let us analyse the natural language inference example mentioned earlier, where the NLP model wrongly predicted, with high confidence, that the premise "John killed Mary" entails the hypothesis "Mary killed John". This is again due to a spurious association that has been learnt by the model from the training data. These two sentences have a high degree of lexical overlap (all the words are common for both the sentences even though the order of appearance is different). In the training set on which the models were trained, a big majority of premise-hypothesis pairs which had a high degree of lexical overlap, were labelled as textual entailment. Hence, the model learnt a spurious short cut heuristic that if the premise and hypothesis sentence pairs have a very high degree of lexical overlap, they are likely to be textual entailment, which is incorrect.

The model used this incorrect heuristic to classify the given example which had 100 per cent overlap, to mark it as textual entailment. As we pointed out before, this error happened because of the underlying associative learning paradigm on which NLP models are built.

Given that NLP models are brittle because of these spurious cues/patterns they have incorrectly picked up from the training data, what can we do to make our models resistant to these shallow statistical patterns? At first glance, the easy solution seems to be adding more data to train the model. For the natural language inference task, we could add more examples, but how do we make sure that the examples we add do not carry the same spurious pattern we have mentioned above? More importantly, how can we add more examples that do NOT have a high degree of lexical overlap but the sentence pair forms an entailment. We also need to add counter examples where there is high lexical overlap between premise and hypothesis, but there is no entailment relationship between them. When we collect data, how can we ensure that such requirements are met? Hence blindly adding more examples to training data may not help, and in certain cases, can make the problem worse. Also, adding training data entails the costs of data collection and labelling.

Instead of just adding new examples to training data, can we somehow augment data samples such that they address the spurious associations we are trying to prevent the model from learning? Or are there alternative methods by which we can still retain the spurious associations in the data, but somehow prevent the model from learning them? These are the topics we will cover in the next column.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Wishing all our readers happy coding until next month! END

By: Sandya Mannarswamy

The author is an expert in natural language processing and is currently working as an independent researcher. Her interests include natural language processing, machine learning and AI.